第十六讲
应力状态分析

作业：8-2(a,c),4(a,c)，5,7,8(b)；

第八章　应力应变状态分析

8.1 平面应力状态应力分析
8.2 应力圆
8.3 极值应力与主应力
8.4 复杂应力状态的最大应力
8.5 平面内应变分析
8.6 广义虎克定律
8.7 复杂应力状态下的应变能
8.8 复合材料的应力应变关系简介（拓展内容）

第八章 应力应变状态分析

8.1 平面应力状态应力分析

1、问题的提出

2、平面与空间应力状态

3、斜截面应力分析——解析法

4、例题

1、问题的提出

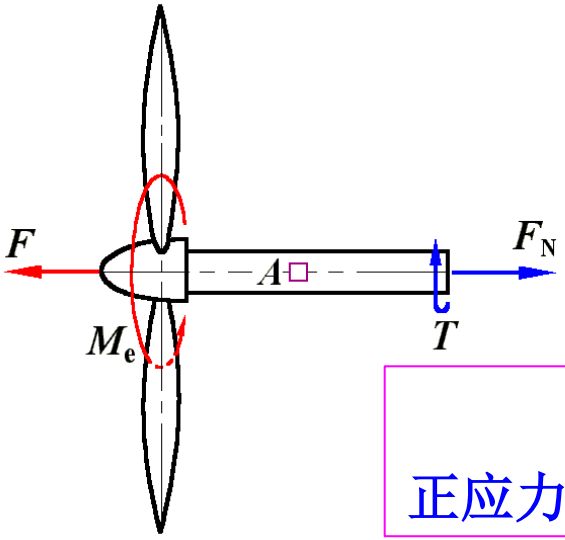
杆件拉压弯扭的强度条件——单一载荷作用

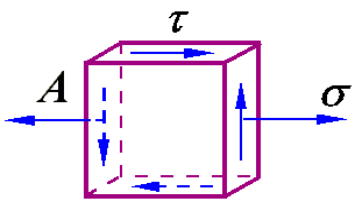
单向受力状态 $\sigma_{\max} \le [\sigma]$

纯剪切应力状态 $\tau_{\max} \le [\tau]$

依据试验建立

其它受力状态的强度条件如何建立？



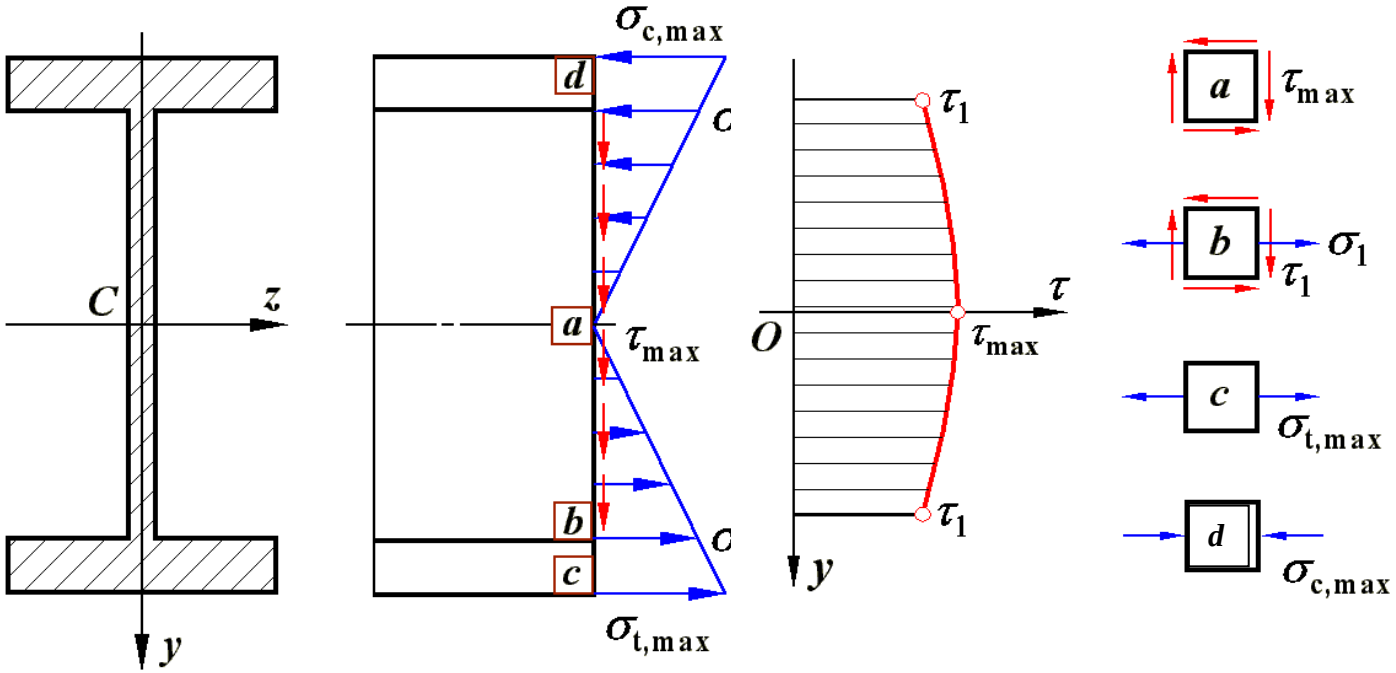


多载荷作用

正应力与切应力彼此独立

1、问题的提出

其它受力状态的强度条件如何建立？



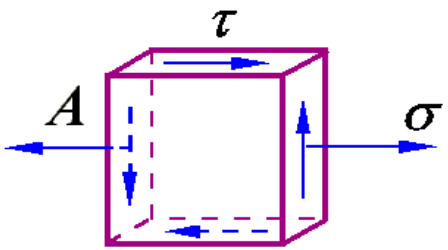
b 点处 - σ,τ 联合作用

多内力作用
正应力与切应力彼此独立

1、问题的提出

其它受力状态的强度条件如何建立？

$\sigma=[\sigma]$，$\tau=[\tau]$ ？ 早破坏了？！



$\sigma=\dfrac{[\sigma]}{2}$，$\tau=\dfrac{[\tau]}{2}$ or，$\dfrac{\sigma}{[\sigma]}+\dfrac{\tau}{[\tau]}=1$？ 没有依据！

试验测试——需指定应力比

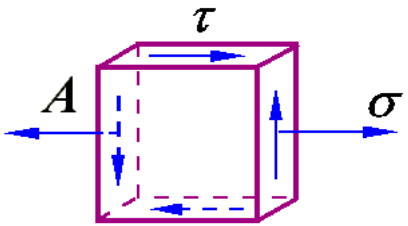
$\sigma=k\tau \Rightarrow [\sigma_k]$

$[\sigma_1]\neq[\sigma_2]\neq\cdots$

- 每一应力比下的许用应力均不相同；
- 应力比无数，需做无数试验！
- 此路不通。

1、问题的提出

其它受力状态的强度条件如何建立？



分析构件的破坏形式

↓

引起破坏的主要因素和条件

强度理论

研究危险点处应力应变状态

↓

最大应力、应变等参量的大小和方位

应力应变状态理论

1、问题的提出

应力状态分析

研究一点处所有微截面上的应力状况，
即正应力和切应力与微截面位置的关系。

应变状态分析

研究一点处所有方位上的应变状况，
即正应变和切应变与所在方位的关系。

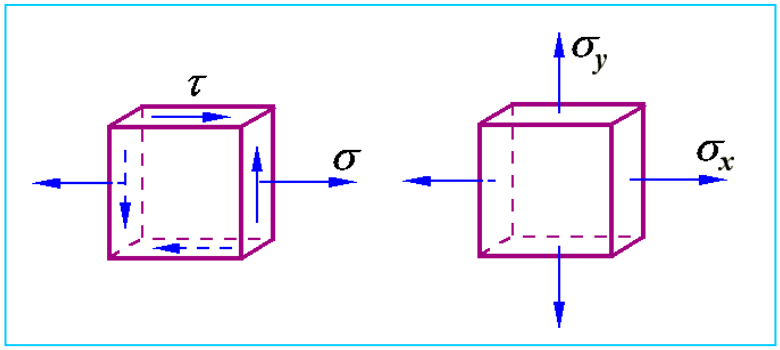
研究目的

最大应力、应变是多少？在何处？
应变和应变的关系。

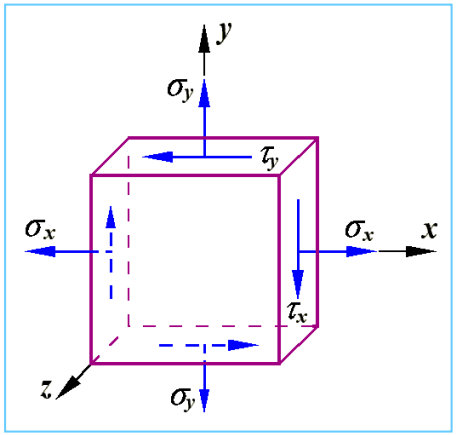
研究方法

切取微体：表示各截面与方位；
应力状态分析：微体静力平衡；
应变状态分析：切线代圆弧。

2、平面与空间应力状态

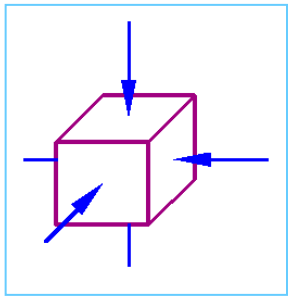


仅在微体四侧面作用应力，且应力作用线均平行于微体的不受力侧面—平面应力状态



平面应力状态的一般形式

$$\sigma_x, \sigma_y, \tau_x = \tau_y$$

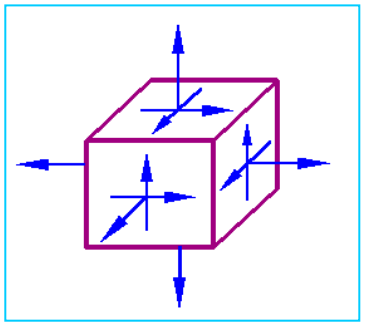


微体各侧面均作用有应力

—空间应力状态

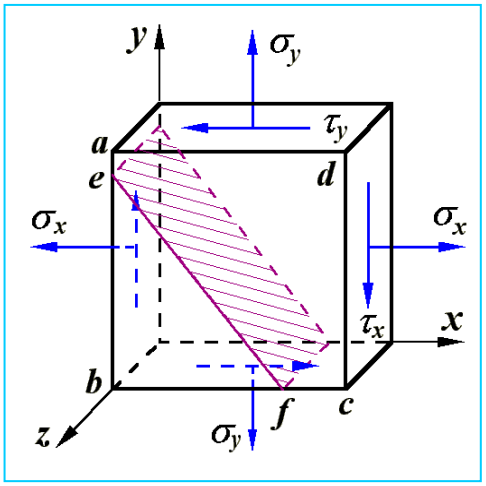
空间应力状态一般形式

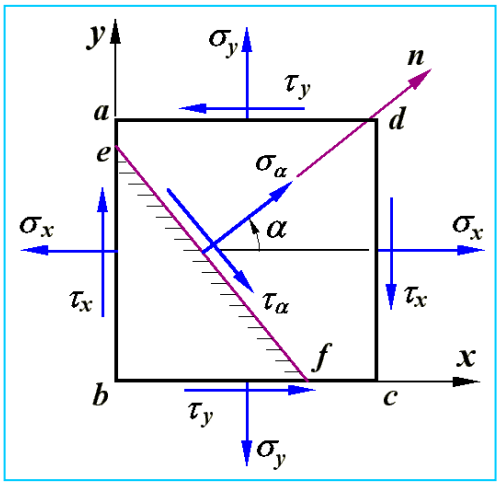
六个应力分量



3、斜截面应力分析——解析法

平面应力状态





任意截面：// z 轴；　截面方位角用 α 表示；应力为 $\sigma_\alpha, \tau_\alpha$

符号规定：❶ 正应力 σ　—以拉为正；

切应力 τ — 以企图使微体沿 ↻ 旋转者为正

❷ 方位角 α — 以 x 轴为始边、↺ 者为正

问题：试建立 $\sigma_\alpha, \tau_\alpha$ 与 $\sigma_x, \sigma_y, \tau_x$间的关系

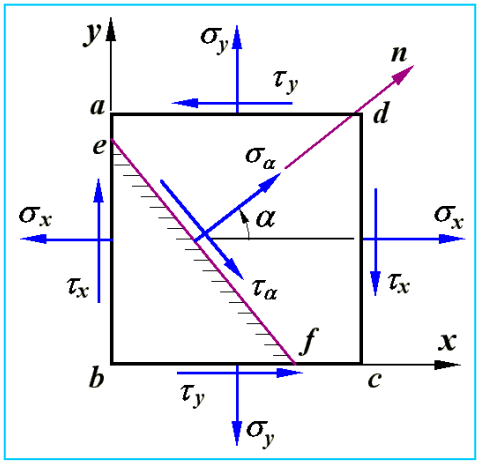
方法：利用微体的平衡条件求解

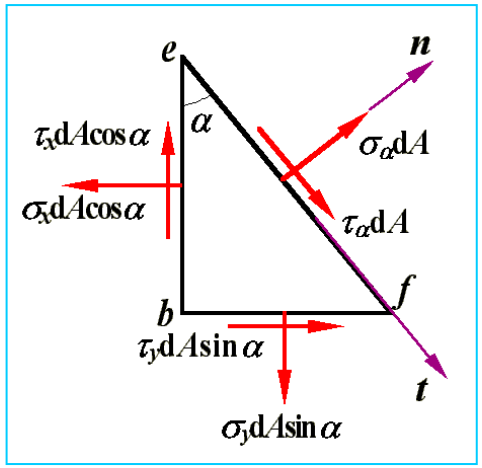
3、斜截面应力分析——解析法

α截面应力公式

根据平衡条件确定

微体应力图





微体受力图

$$\Sigma F_n = 0,\quad \sigma_\alpha dA + (\tau_x dA\cos\alpha)\sin\alpha - (\sigma_x dA\cos\alpha)\cos\alpha + (\tau_y dA\sin\alpha)\cos\alpha - (\sigma_y dA\sin\alpha)\sin\alpha = 0$$

$$\Sigma F_t = 0,\quad \tau_\alpha dA - (\tau_x dA\cos\alpha)\cos\alpha - (\sigma_x dA\cos\alpha)\sin\alpha + (\tau_y dA\sin\alpha)\sin\alpha + (\sigma_y dA\sin\alpha)\cos\alpha = 0$$

$$\sigma_\alpha = \sigma_x\cos^2\alpha + \sigma_y\sin^2\alpha - (\tau_x + \tau_y)\sin\alpha\cos\alpha$$

$$\tau_\alpha = (\sigma_x - \sigma_y)\sin\alpha\cos\alpha + \tau_x\cos^2\alpha - \tau_y\sin^2\alpha$$

3、斜截面应力分析——解析法

$$\sigma_\alpha = \sigma_x \cos^2\alpha + \sigma_y \sin^2\alpha - (\tau_x + \tau_y)\sin\alpha\cos\alpha$$

$$\tau_\alpha = (\sigma_x - \sigma_y)\sin\alpha\cos\alpha + \tau_x \cos^2\alpha - \tau_y \sin^2\alpha$$

由于τ_x与τ_y数值相等，并利用三角函数的变换关系，得

应力转轴公式

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_x \sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_x \cos 2\alpha$$

上述关系式是建立在静力学基础上，因而所得结论既适用于各向同性与线弹性情况，也适用于各向异性、非线弹性与非弹性问题

3、斜截面应力分析——解析法

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_x \sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_x \cos 2\alpha$$

讨论：

$\alpha = 0,$ x面应力状态；

$\alpha = \frac{\pi}{2},$ y面应力状态

$\sigma_y = 0, \tau_x = 0,$ ⟶ 拉压杆斜截面应力（式(2-2),(2-3)）；

$$\sigma_\alpha = \frac{\sigma_x}{2}(1+\cos 2\alpha) = \sigma_x \cos^2\alpha, \quad \tau_\alpha = \frac{\sigma_x}{2}\sin 2\alpha \qquad \alpha = \pm\frac{\pi}{4}, \quad \tau_{\max} = \pm\frac{\sigma_x}{2}$$

$\sigma_x = \sigma_y = 0,$ ⟶ 受扭圆轴斜截面应力；

$$\sigma_\alpha = -\tau_x \sin 2\alpha, \quad \tau_\alpha = \tau_x \cos 2\alpha \qquad \alpha = \frac{\pi}{4}, \quad \sigma_{45} = -\tau_x, \quad \tau_{45} = 0;$$

$$\alpha = -\frac{\pi}{4}, \quad \sigma_{-45} = \tau_x, \quad \tau_{-45} = 0$$

3、斜截面应力分析——解析法

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_x \sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_x \cos 2\alpha$$

说明：

1）符号规定：

+σ：受拉；+τ：顺时针；

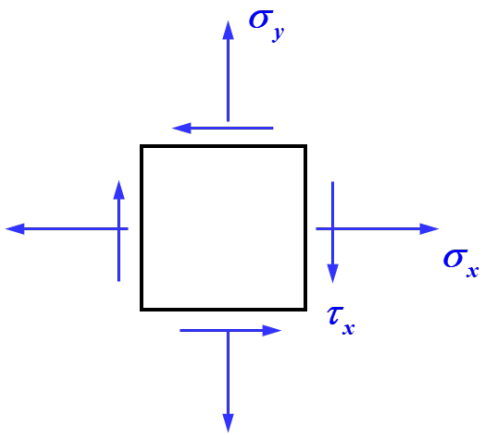
+α：+x逆时针。

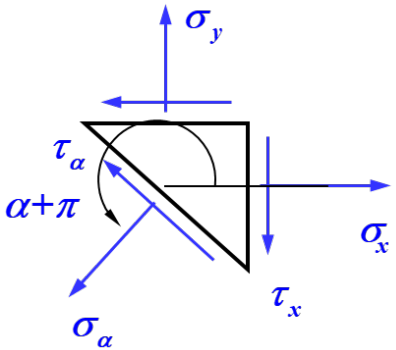
2）α面与α+π面：

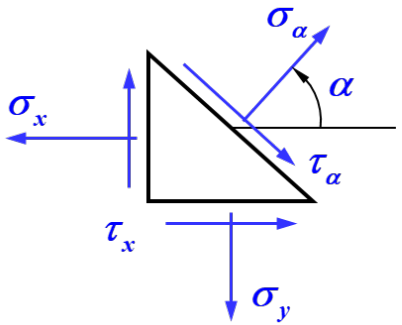
$$\sigma_\alpha = \sigma_{\alpha+\pi}, \quad \tau_\alpha = \tau_{\alpha+\pi}$$

3）应力公式 → 应力极值、方位角

顶顶





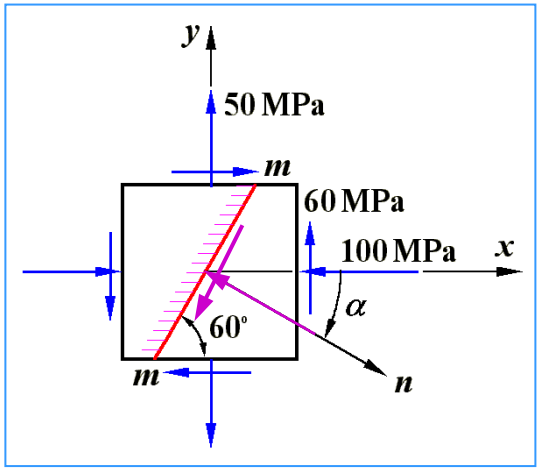


4、例题

例8-1　试计算截面 *m-m* 上的应力

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_x \sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_x \cos 2\alpha$$



解： $\sigma_x = -100\ \text{MPa}$　$\tau_x = -60\ \text{MPa}$　$\sigma_y = 50\ \text{MPa}$　$\alpha = -30^\circ$

$$\sigma_m = \frac{-100+50}{2} + \frac{-100-50}{2}\cos(-60^\circ) - (-60)\sin(-60^\circ) = -114.5\ \text{MPa}$$

$$\tau_m = \frac{-100-50}{2}\sin(-60^\circ) + (-60)\cos(-60^\circ) = 35.0\ \text{MPa}$$

第八章 应力应变状态分析

8.2 应力圆

1、张行教授画的应力圆

2、应力圆的绘制

3、应力圆的特点

4、例题

5、赵汶鑫旋转向量画应力圆

1、张行教授画的应力圆

王俊奎教授

张行教授

1、张行教授画的应力圆

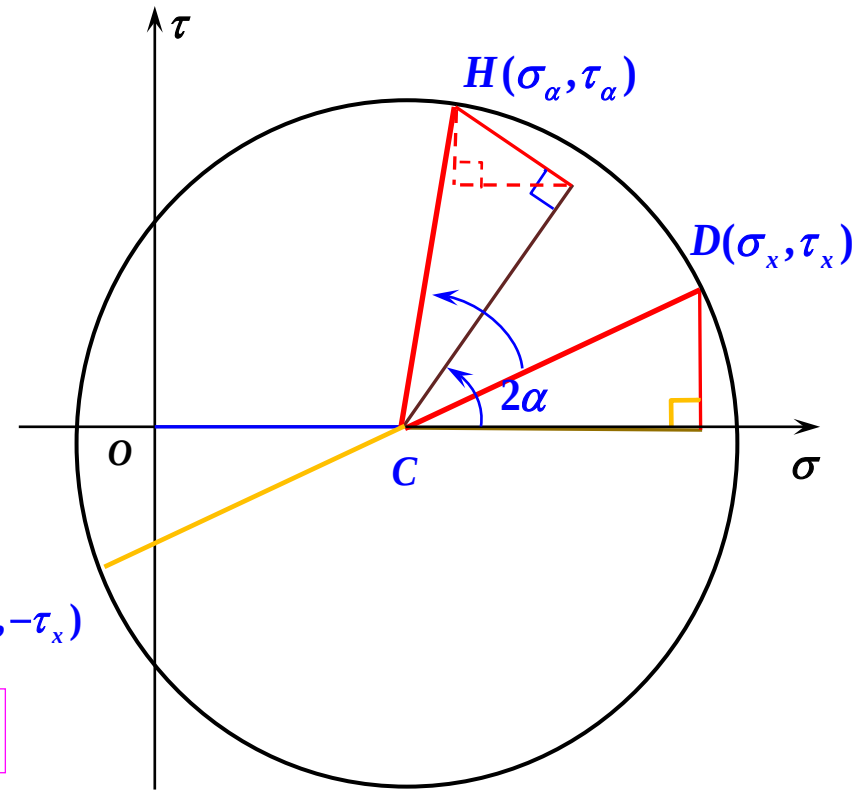
斜截面应力分析的图解法

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_x \sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_x \cos 2\alpha$$

$$\frac{\sigma_x + \sigma_y}{2} \qquad \frac{\sigma_x - \sigma_y}{2} \qquad \tau_x$$

应力圆
（Mohr圆）



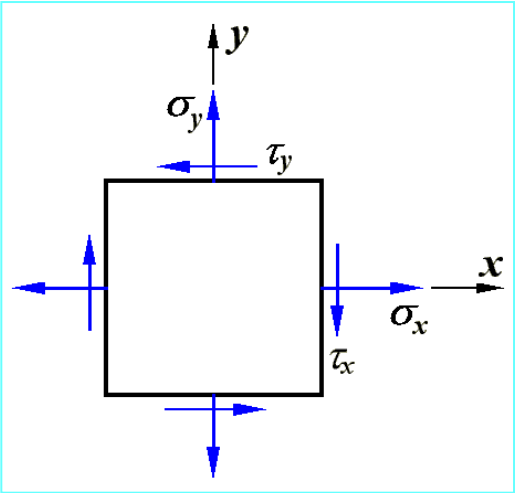
σ-τ坐标系中点 $(\sigma_\alpha, \tau_\alpha)$ ⟷ α截面应力

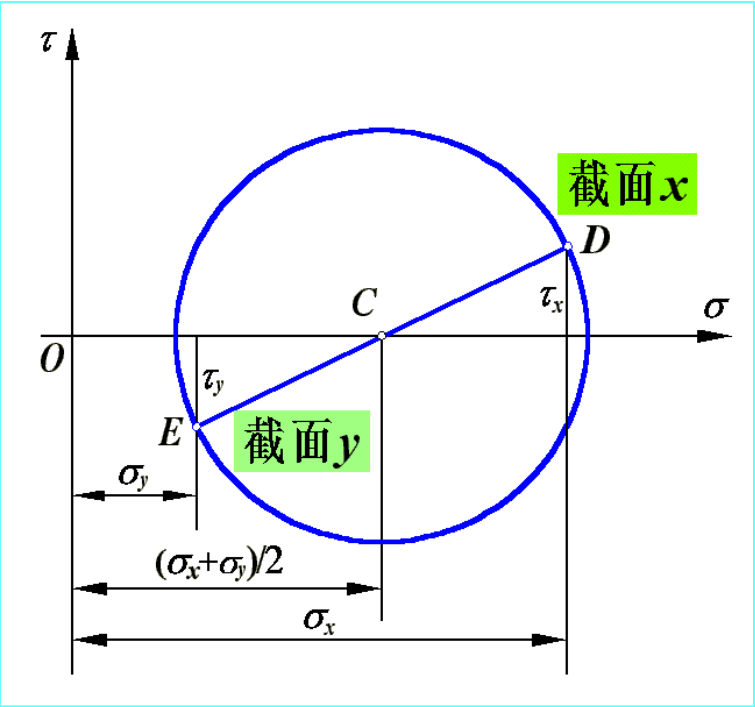
应力圆 ⇨ 应力状态

$$C\left(\frac{\sigma_x + \sigma_y}{2}, 0\right), \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_x^2}$$

图解法，直观、简洁

2、应力圆的绘制





x面应力

y面应力

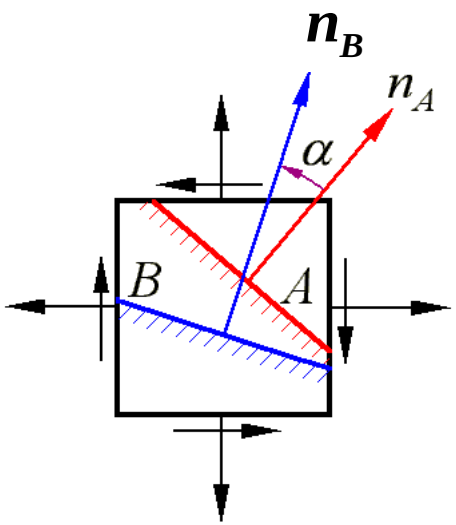
D、E两点

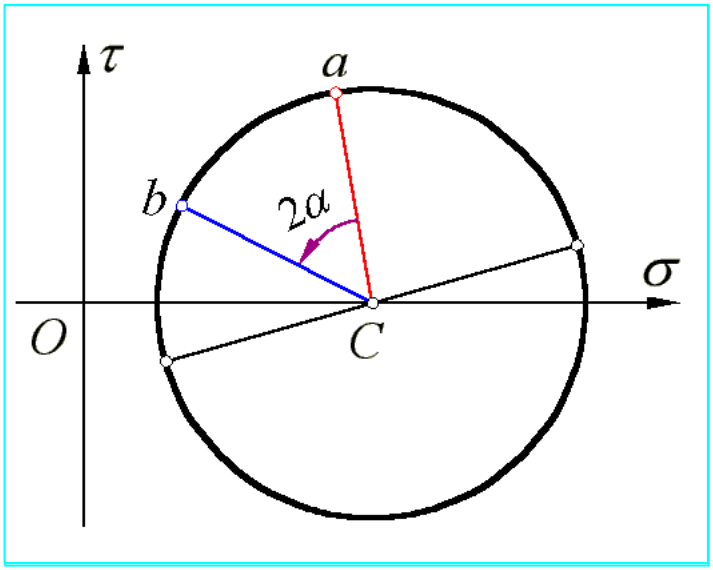
$$\sigma_C = \frac{\sigma_x + \sigma_y}{2}$$ — 圆心横坐标

应力圆直径DE → 应力圆

3、应力圆的特点

点、面对应关系

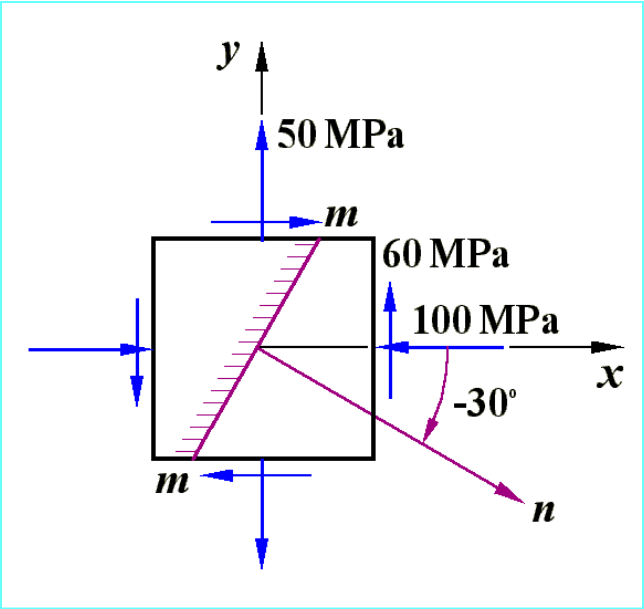


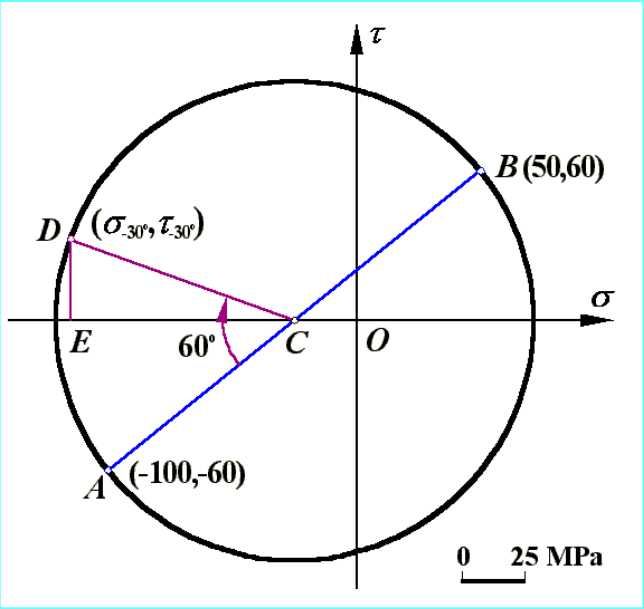


- 点面对应，转向相同，转角加倍；
- 圆心在σ轴上；
- 互垂截面，对应同一直径两端。

4、例题

例8-2 利用应力圆求截面 *m-m* 上的应力





解：

$$\sigma_m = -115\ \text{MPa} \qquad \tau_m = 35\ \text{MPa}$$

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5、赵汶鑫旋转向量画应力圆

定义向量

x面向量

$$\overrightarrow{OD}=\begin{Bmatrix}\sigma_x\\ \tau_x\end{Bmatrix},\quad \overrightarrow{OC}=\begin{Bmatrix}\dfrac{\sigma_x+\sigma_y}{2}\\ 0\end{Bmatrix}$$

可得向量

x面半径

$$\overrightarrow{CD}=\overrightarrow{OD}-\overrightarrow{OC}=\begin{Bmatrix}\sigma_x\\ \tau_x\end{Bmatrix}-\begin{Bmatrix}\dfrac{\sigma_x+\sigma_y}{2}\\ 0\end{Bmatrix}=\begin{Bmatrix}\dfrac{\sigma_x-\sigma_y}{2}\\ \tau_x\end{Bmatrix}$$

旋转得到

α面半径

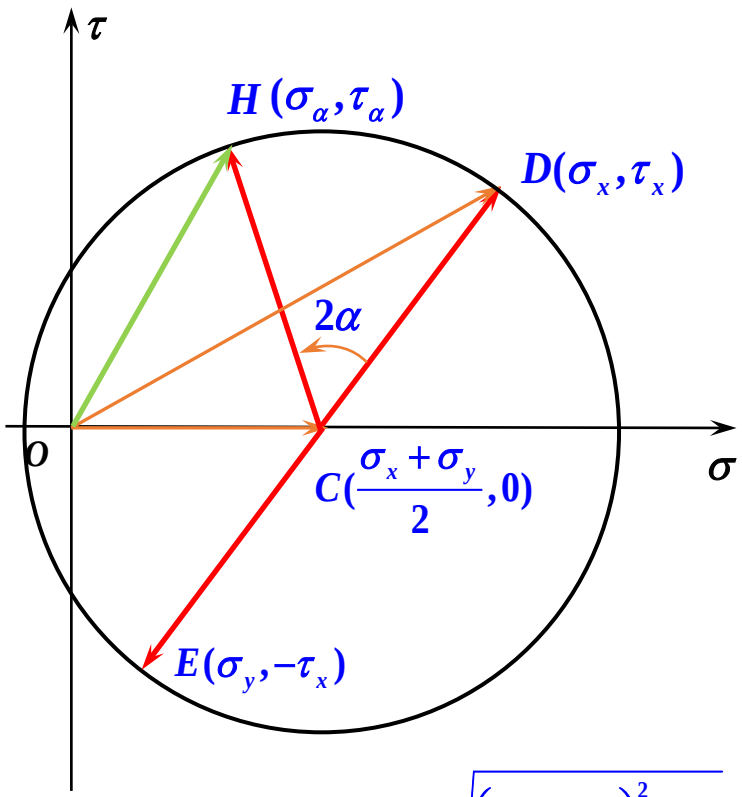
$$\overrightarrow{CH}=\begin{bmatrix}\cos2\alpha & -\sin2\alpha\\ \sin2\alpha & \cos2\alpha\end{bmatrix}\begin{Bmatrix}\dfrac{\sigma_x-\sigma_y}{2}\\ \tau_x\end{Bmatrix}$$

α面向量

可见

$$\overrightarrow{OH}=\overrightarrow{OC}+\overrightarrow{CH}$$

$$=\begin{Bmatrix}\dfrac{\sigma_x+\sigma_y}{2}\\ 0\end{Bmatrix}+\begin{bmatrix}\cos2\alpha & -\sin2\alpha\\ \sin2\alpha & \cos2\alpha\end{bmatrix}\begin{Bmatrix}\dfrac{\sigma_x-\sigma_y}{2}\\ \tau_x\end{Bmatrix}=\begin{Bmatrix}\sigma_\alpha\\ \tau_\alpha\end{Bmatrix}$$



$$C(\frac{\sigma_x+\sigma_y}{2},0),\quad R=\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_x^2}$$

向前一步，
你会看到一个新的世界！

第八章 应力应变状态分析

8.3 平面应力状态的极值应力与主应力

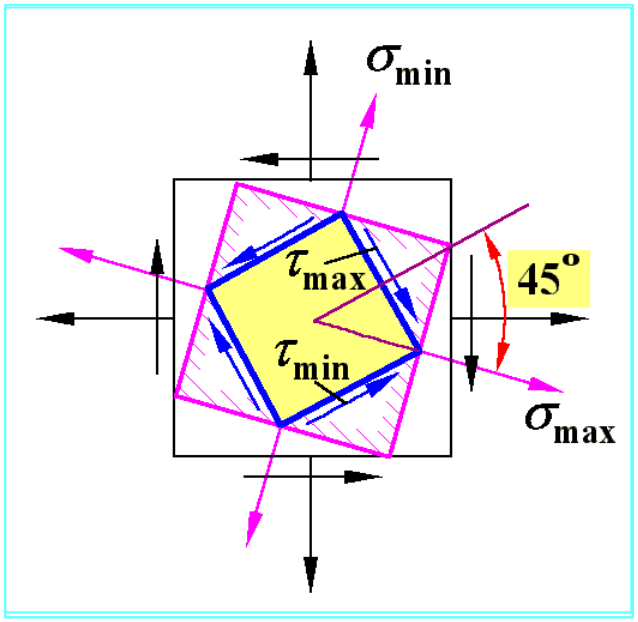
1、平面应力状态的极值应力

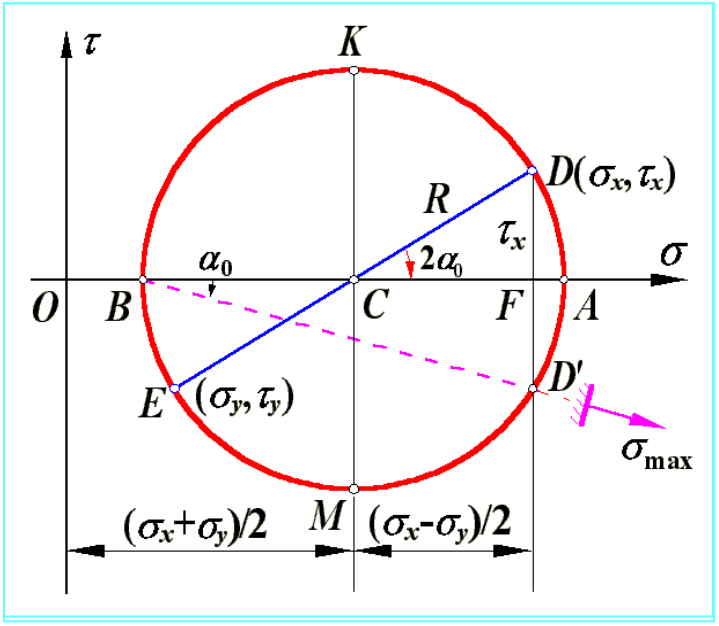
2、主平面与主应力

3、纯剪切应力与扭转破坏

4、例题

1、平面应力状态的极值应力





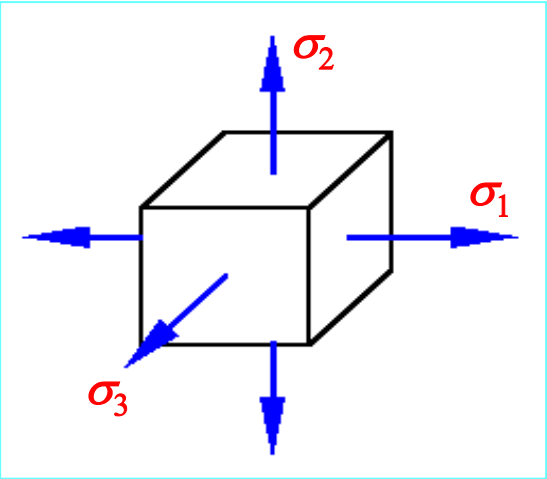
$$\begin{matrix}\sigma_{\max}\\ \sigma_{\min}\end{matrix}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_x^2}$$

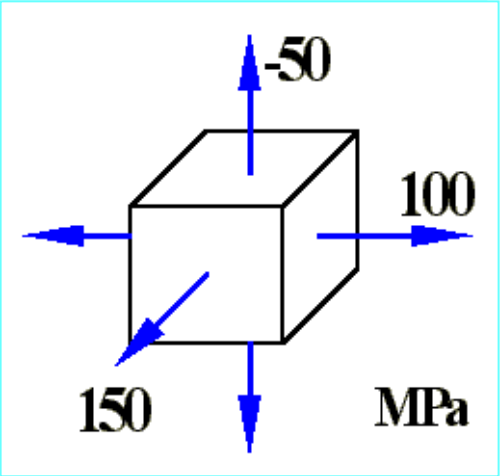
$$\begin{matrix}\tau_{\max}\\ \tau_{\min}\end{matrix}=\pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_x^2}$$

$$\tan 2\alpha_0=-\frac{2\tau_x}{\sigma_x-\sigma_y}$$

$$\tan\alpha_0=-\frac{\tau_x}{\sigma_x-\sigma_{\min}}=-\frac{\tau_x}{\sigma_{\max}-\sigma_y}$$

2、主平面与主应力





$\sigma_i = ?$

主平面－切应力为零的截面

相邻主平面相互垂直，构成一六面体微体

——主平面微体

主应力－主平面上的正应力

主应力符号与规定－ $\sigma_1 \geq \sigma_2 \geq \sigma_3$（按代数值排列）

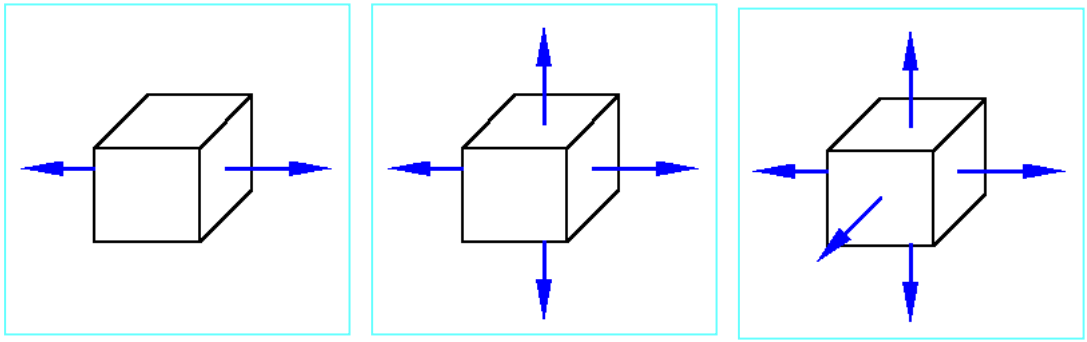
2、主平面与主应力

应力状态分类

❶ 单向应力状态：仅一个主应力不为零的应力状态

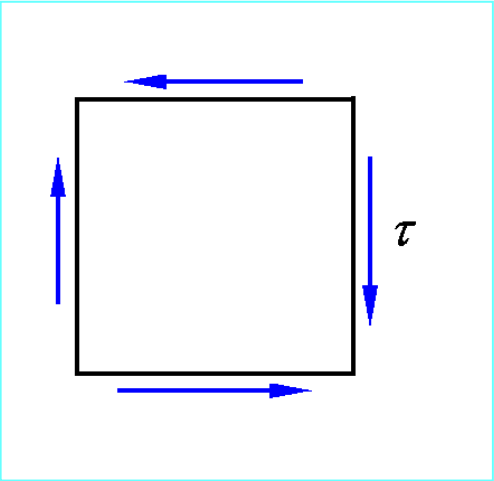
❷ 二向应力状态：两个主应力不为零的应力状态

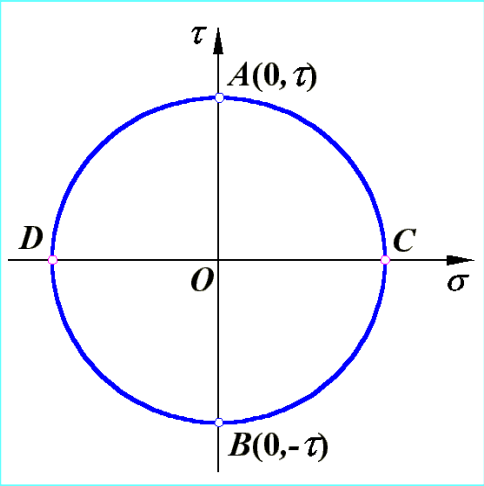
❸ 三向应力状态：三个主应力均不为零的应力状态

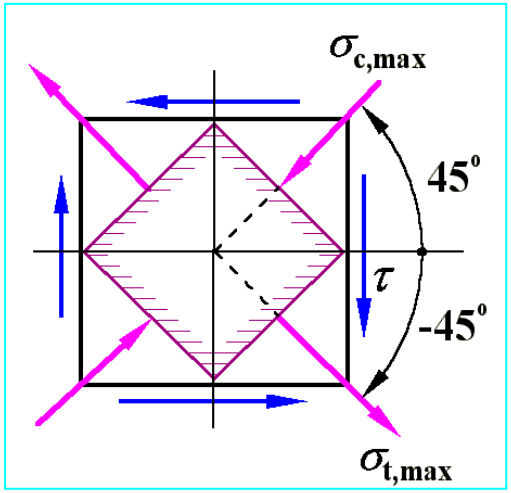


二向与三向应力状态，统称复杂应力状态

3、纯剪切应力与扭转破坏



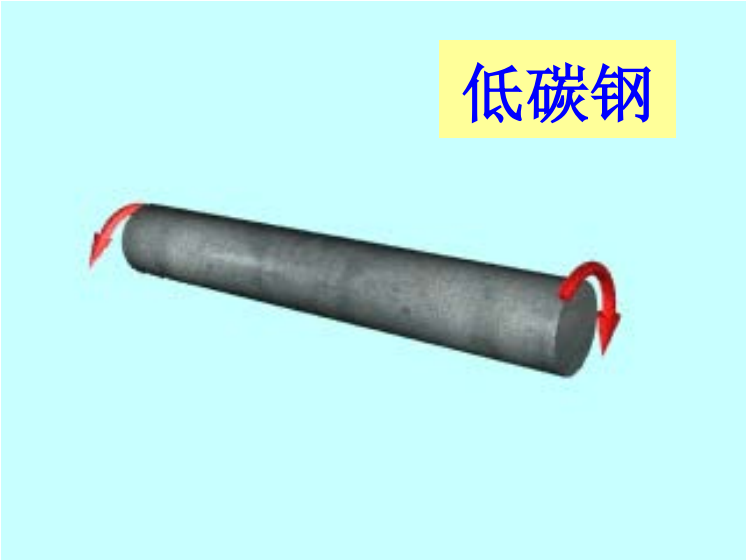


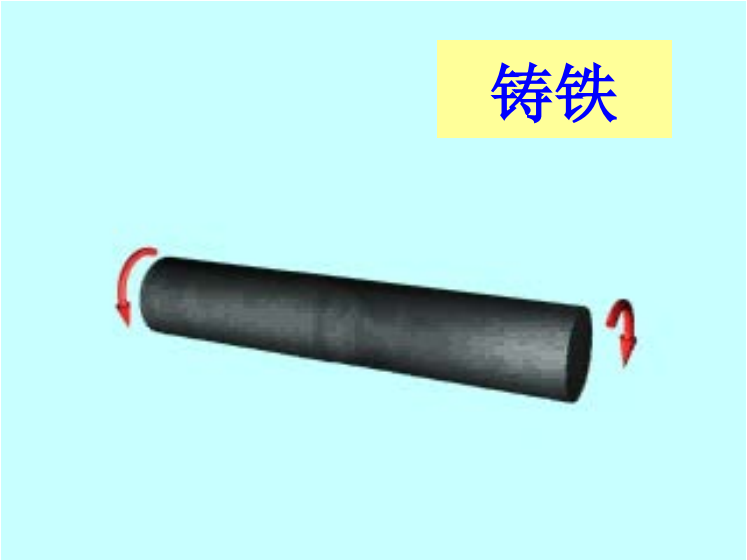


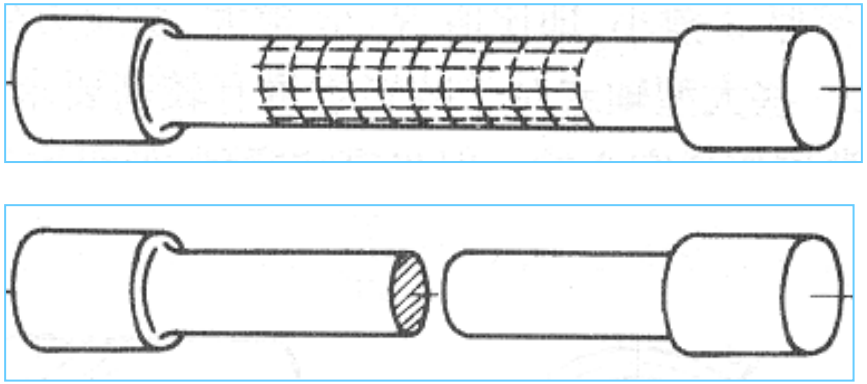
$$\sigma_{\mathrm{t,max}} = \sigma_c = \tau \qquad \tau_{\max} = -\tau_{\min} = \tau$$

$$\sigma_{\mathrm{c,max}} = |\sigma_D| = \tau \qquad \sigma_1 = -\sigma_3 = \tau, \quad \sigma_2 = 0$$

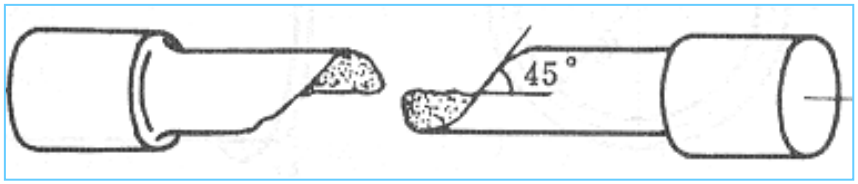
3、纯剪切应力与扭转破坏







滑移与剪断发生在τ_{max}的作用面



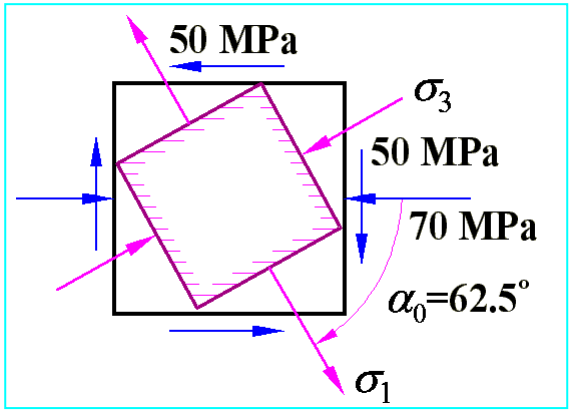
断裂发生在σ_{max}的作用面

4、例题

例8-3　用解析法与图解法，确定主应力的大小与方位

$$\begin{matrix}\sigma_{\max}\\ \sigma_{\min}\end{matrix}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)+\tau_x^2}$$

$$\tan\alpha_0=-\frac{\tau_x}{\sigma_{\max}-\sigma_y}$$



解： 1. 解析法　$\sigma_x=-70\,\text{MPa}$　$\tau_x=50\,\text{MPa}$　$\sigma_y=0$

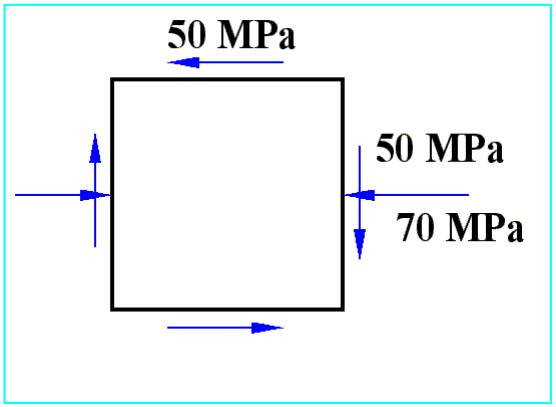
$$\begin{matrix}\sigma_{\max}\\ \sigma_{\min}\end{matrix}=\frac{-70+0}{2}\pm\sqrt{\left(\frac{-70-0}{2}\right)^2+50^2}=\begin{matrix}26\,\text{MPa}\\ -96\,\text{MPa}\end{matrix}$$

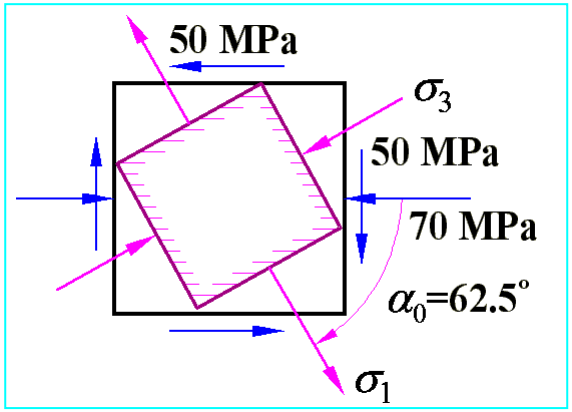
$$\alpha_0=\arctan\left(-\frac{50}{26-0}\right)=-62.5^\circ$$

$$\sigma_1=26\,\text{MPa}\qquad\sigma_2=0\qquad\sigma_3=-96\,\text{MPa}$$

4、例题

例8-3　用解析法与图解法，确定主应力的大小与方位





解： 1. 解析法

$$\alpha_0 = \arctan\left(-\frac{50}{26-0}\right) = -62.5^\circ$$

若按公式　$\tan 2\alpha_0 = -\dfrac{2\tau_x}{\sigma_x - \sigma_y}$

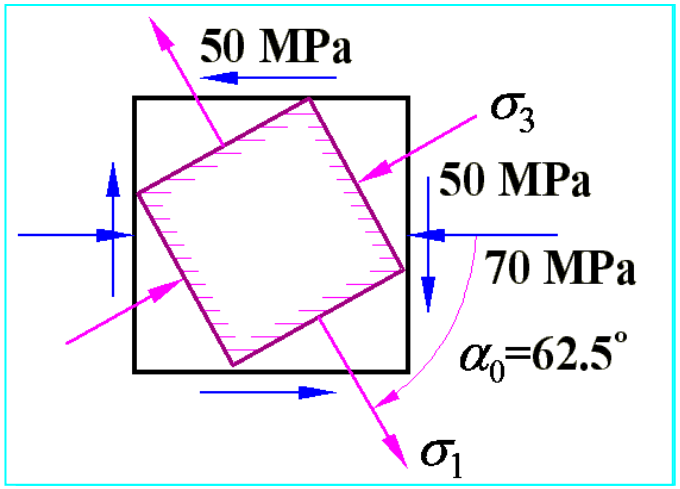
$$\alpha_0 = \frac{1}{2}\arctan\left(-\frac{2\times 50}{-70-0}\right) = \frac{1}{2}\times 55.0^\circ = 27.5^\circ$$

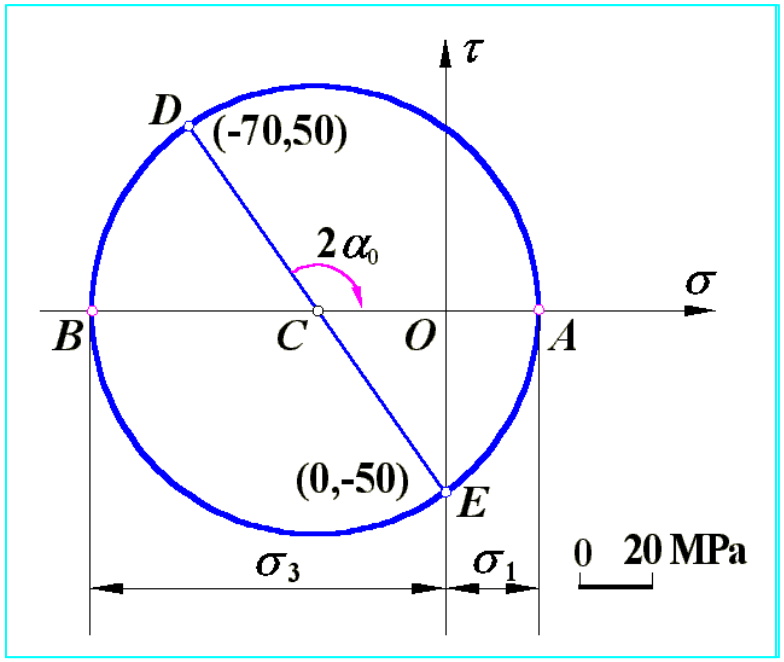
为什么？

4、例题

例8-3　用解析法与图解法，确定主应力的大小与方位

2. 图解法





$$\alpha_0 = -62.5^\circ$$

$$\sigma_1 = 26\ \text{MPa} \qquad \sigma_2 = 0 \qquad \sigma_3 = -96\ \text{MPa}$$

第八章 应力应变状态分析

8.4 复杂应力状态的最大应力

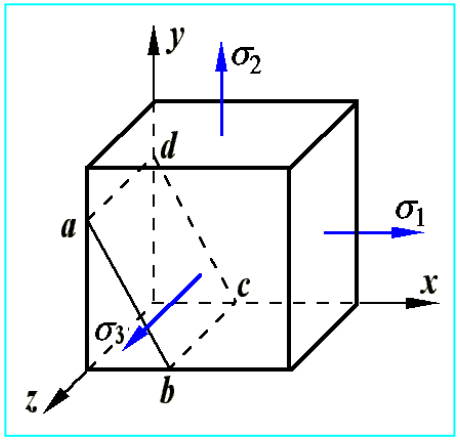
1、三向应力圆

2、最大应力

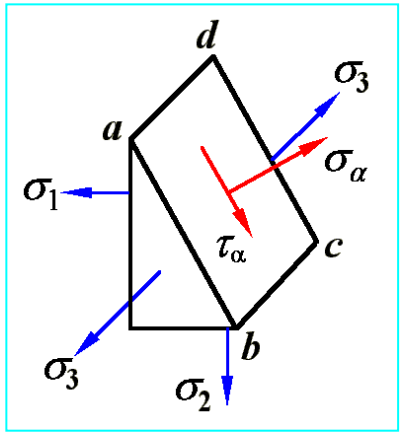
3、例题

1、三向应力圆

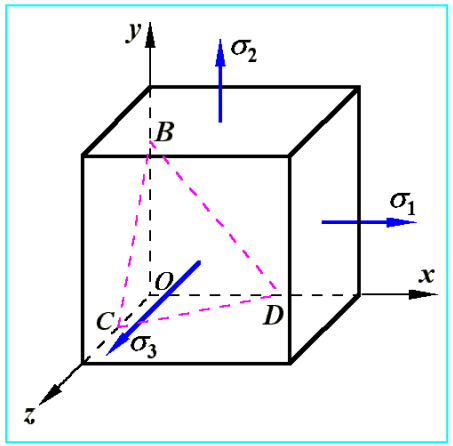
以主应力表示的应力状态



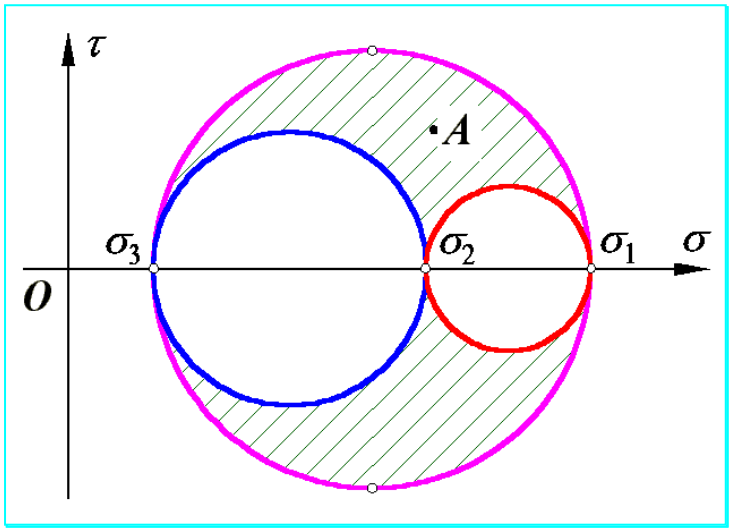
平行于主应力的斜截面



任意斜截面

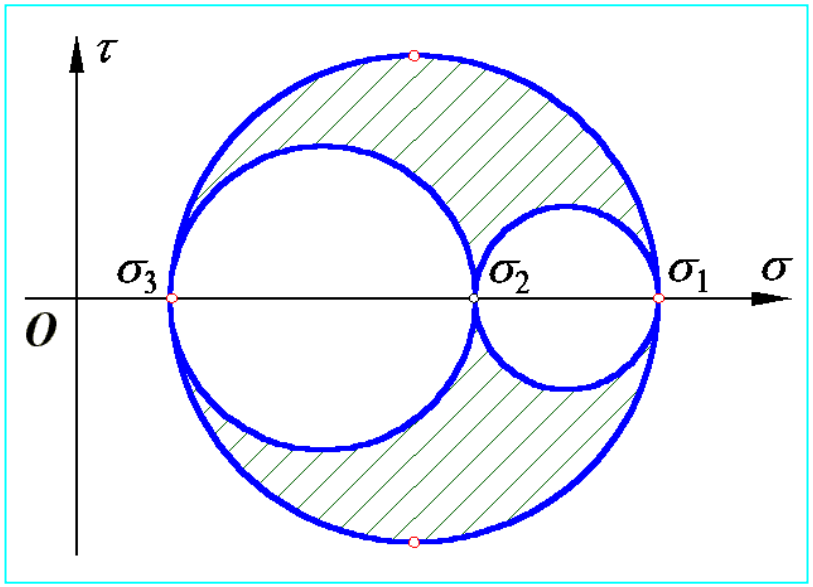


三向应力圆



与任一截面相对应的点，或位于应力圆上，
或位于由应力圆所构成的阴影区域内。（与三主应力相关）
（证明见题8-10（第3版））

2、最大应力



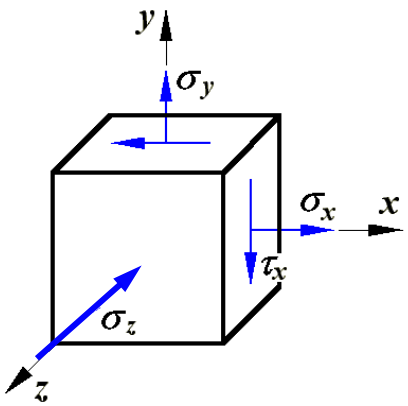
$$\sigma_{\max} = \sigma_1 \qquad \sigma_{\min} = \sigma_3$$

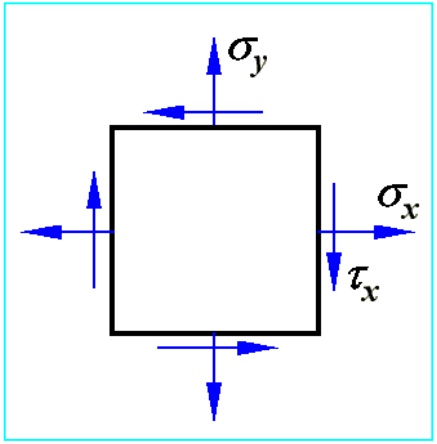
$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2}$$

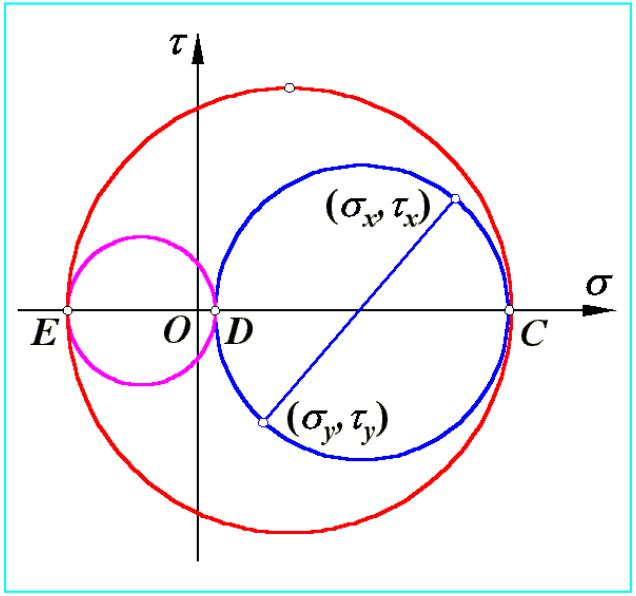
最大切应力位于与 σ_1 及 σ_3 均成45°的截面

3、例题

例8-4　已知：$\sigma_x = 80$ MPa，$\tau_x = 35$ MPa，$\sigma_y = 20$ MPa，$\sigma_z = -40$ MPa，求主应力、最大正应力和切应力。







解：　$\sigma_E = -40$ MPa 是主应力之一

画三向应力圆

$$\sigma_1 = \sigma_C = 96.1\,\text{MPa} \quad \sigma_2 = \sigma_D = 3.09\,\text{MPa} \quad \sigma_3 = \sigma_E = -40\,\text{MPa}$$

$$\sigma_{\max} = \sigma_1 = 96.1\,\text{MPa} \qquad \tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = 68.1\,\text{MPa}$$

本讲：应力状态分析

点面对应、转向相同、转角加倍

下讲：应变状态分析